Problem: Search for D Intervals that Result in Equal SNR

Approach 1: Distribute Decay Curves Evenly

Decay curve

$$S(x) = S_0 e^{(-xA)}$$

Point at which a decay curve reaches S_T

$$x_A = -\frac{\ln \frac{S_T}{S_0}}{A}$$

Distance between two curves with diffusion coefficient A and $A + \Delta A$ at level S_T should be constant:

$$x_{A-\Delta A} - x_A = -\left(\frac{1}{A - \Delta A} - \frac{1}{A}\right) \ln \frac{S_T}{S_0} = \text{const}$$

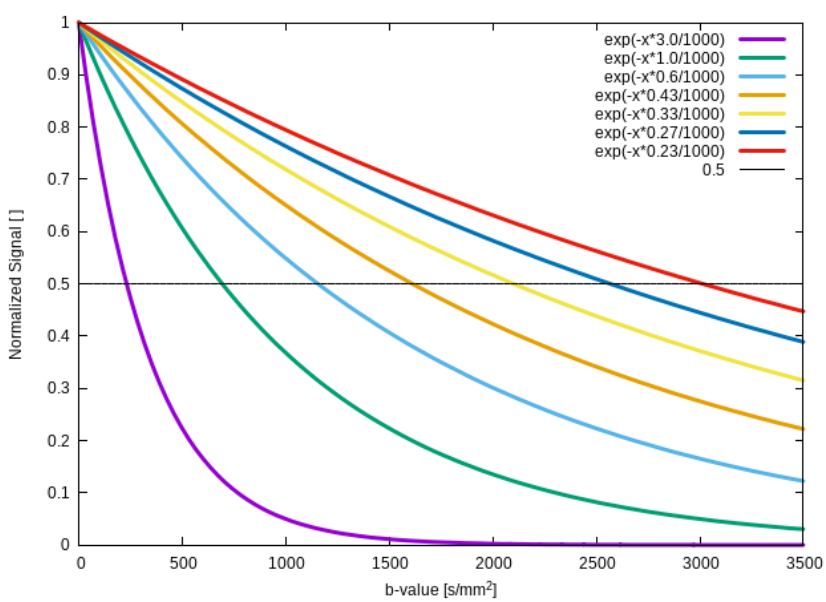
$$\frac{1}{A - \Delta A} - \frac{1}{A} = \frac{1}{K}$$

$$\Delta A = \frac{A^2}{K + A}$$

For A=3.00x10^{-3} and K=1.5x10^{-3} we obtain:

{3.00, 1.00, 0.60, 0.43, 0.33, 0.27, 0.23} (10^{-3})

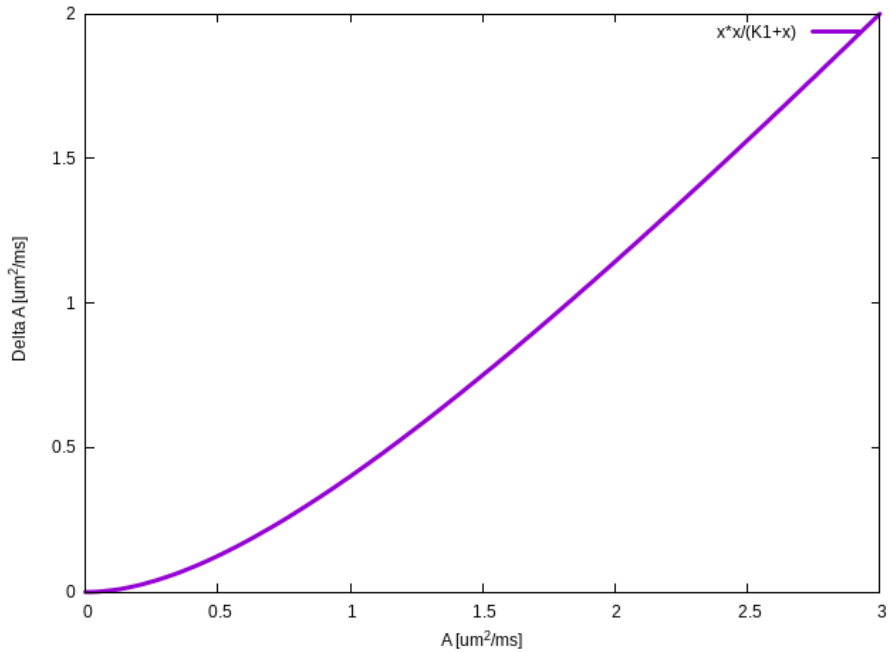
This looks as shown in this plot:



Originally we used:

{3.00, 2.00, 1.50, 1.00, 0.75, 0.50, 0.25} (10^{-3})

Plot of $\Delta A(A)$ with K=1.5



If we start with the lower diffusion coefficient and add ΔA (K remains unchanged):

$$x_A - x_{A+\Delta A} = -\left(\frac{1}{A} - \frac{1}{A+\Delta A}\right)\ln\frac{S_T}{S_0} = \text{const}$$

$$\frac{1}{A} - \frac{1}{A+\Delta A} = \frac{1}{K}$$

$$\Delta A = \frac{A^2}{K-A}$$

Approach 2: CRLB f(S, p)

For the function $f(x) = S \cdot \left[pe^{-Ax} + (1-p)e^{-(A+dA)x}\right]$, the Cramér-Rao Lower Bound (CRLB) for the parameter p given N equally sampled points over an interval $[0, T]$ with Gaussian noise of variance σ^2 is:

$$\text{CRLB}(p) \approx \frac{\sigma^2}{N}\left[\frac{1}{T}\int_0^T \left(S\left(e^{-Ax} - e^{-(A+dA)x}\right)\right)^2 dx\right]^{-1}$$

1. Derive the Fisher Information

The CRLB for a parameter θ is the inverse of the Fisher Information $I(\theta)$. For a signal $f(x)$ corrupted by i.i.d. Gaussian noise $\mathcal{N}(0, \sigma^2)$, the Fisher Information for p is:

$$I(p) = \frac{1}{\sigma^2} \sum_{i=1}^{N} \left(\frac{\partial f(x_i)}{\partial p} \right)^2$$

Calculating the partial derivative:

$$\frac{\partial f}{\partial p} = S \left(e^{-Ax} - e^{-(A+dA)x} \right)$$

For large N, we approximate the sum with an integral over the interval T:

$$I(p) \approx \frac{N}{T\sigma^2} \int_0^T S^2 \left(e^{-Ax} - e^{-(A+dA)x} \right)^2 dx$$

2. Evaluate the Integral

Expanding the square term $\left(e^{-Ax} - e^{-(A+dA)x} \right)^2 = e^{-2Ax} - 2e^{-(2A+dA)x} + e^{-2(A+dA)x}$ and integrating gives:

$$\int_0^T (\ldots) dx = \left[\frac{1 - e^{-2AT}}{2A} - \frac{2(1 - e^{-(2A+dA)T})}{2A + dA} + \frac{1 - e^{-2(A+dA)T}}{2(A + dA)} \right]$$

The CRLB is then:

$$\mathrm{CRLB}(p) = \frac{\sigma^2 T}{NS^2} \left[\frac{1 - e^{-2AT}}{2A} - \frac{2(1 - e^{-(2A+dA)T})}{2A + dA} + \frac{1 - e^{-2(A+dA)T}}{2(A + dA)} \right]^{-1}$$

3. Solve for Constant CRLB

To find $dA(A)$ such that the CRLB remains constant, we examine the sensitivity term in the brackets. If T is large ($AT \gg 1$), the exponential terms vanish, simplifying the information to:

$$I \propto \frac{1}{2A} - \frac{2}{2A + dA} + \frac{1}{2(A + dA)}$$

Re term exp(-*2AT*)<<1:

T=3500 (b-value range)

A=3.00x10^{-3} exp(-*2AT*)=7.5x10^{-10}

A=0.50x10^{-3} exp(-*2AT*)=0.030

A=0.25x10^{-3} exp(-*2AT*)=0.173

$$I \propto \frac{dA^2}{2A(A+dA)(2A+dA)} = K$$

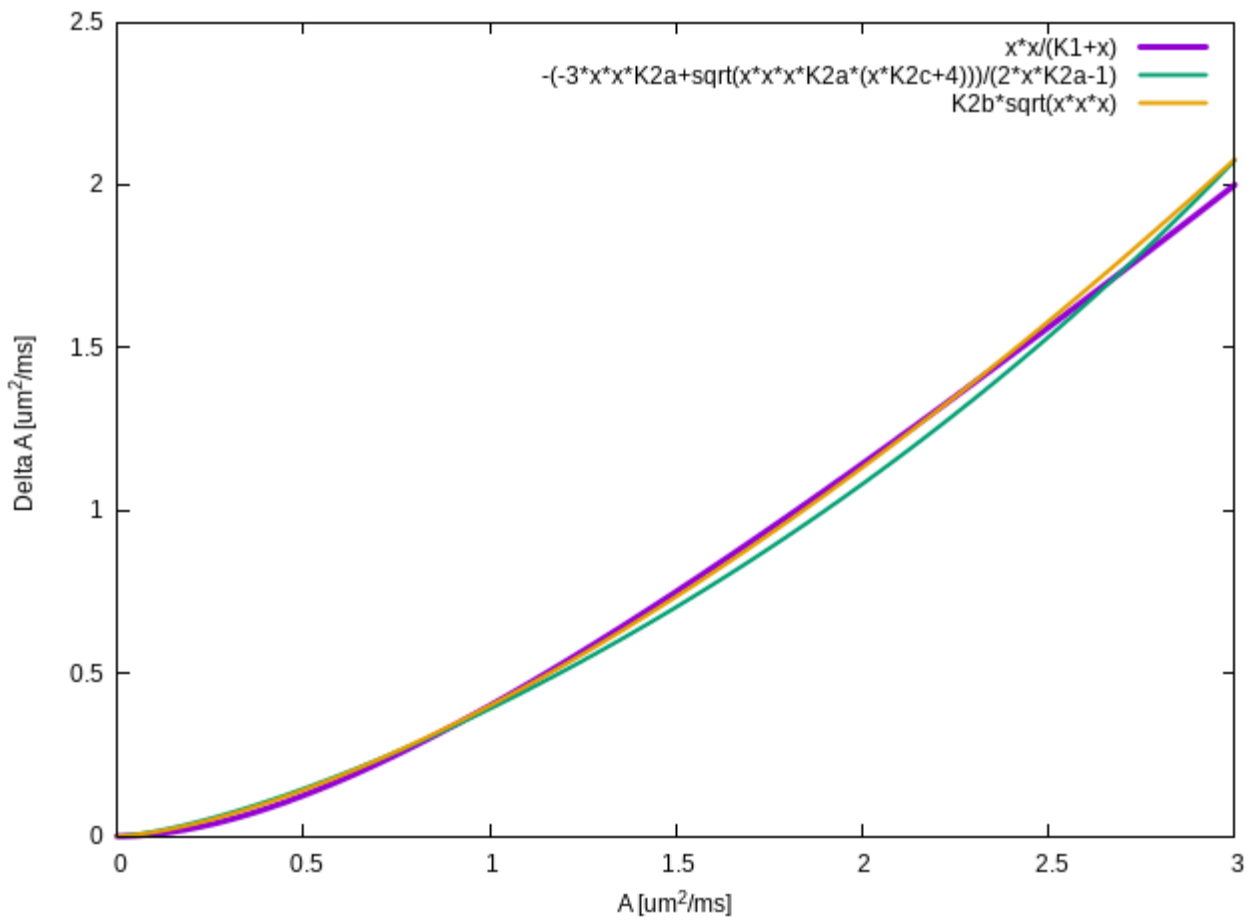
$$dA = \frac{-3A^2K \pm \sqrt{A^3K(AK+4)}}{2AK-1}$$

Or simplified solution (*dA*<<*A*)

$$I \propto \frac{dA^2}{4A^3}$$

$$dA \approx \pm KA^{3/2}$$

Plot of $\Delta A(A)$ for solution 1 (equal interval), 2a (CRLB, for *dA<0*), and 2b (CRLB simplified)



Approach 3: CRLB f(p, q)

For a signal $f(x) = pe^{-Ax} + qe^{-(A+\Delta A)x}$ sampled at N points $x_i = i \cdot dx$ (where $i = 0, 1, \ldots, N-1$) under additive white Gaussian noise with variance σ^2, the **Cramér-Rao Lower Bound (CRLB)** for the amplitudes p and q is given by the diagonal elements of the inverse Fisher Information Matrix (FIM).

1. Derive the Fisher Information Matrix

For Gaussian noise, the FIM entries I_{jk} for parameters θ_j, θ_k are calculated as:

$$I_{jk} = \frac{1}{\sigma^2} \sum_{i=0}^{N-1} \frac{\partial f(x_i)}{\partial \theta_j} \frac{\partial f(x_i)}{\partial \theta_k}$$

The partial derivatives with respect to p and q are:

- $\frac{\partial f}{\partial p} = e^{-Ax_i}$
- $\frac{\partial f}{\partial q} = e^{-(A+\Delta A)x_i}$

The FIM $\mathbf{I}$ is a 2×2 matrix:

$$\mathbf{I} = \frac{1}{\sigma^2} \begin{pmatrix} \sum e^{-2Ax_i} & \sum e^{-(2A+\Delta A)x_i} \\ \sum e^{-(2A+\Delta A)x_i} & \sum e^{-2(A+\Delta A)x_i} \end{pmatrix}$$

2. Calculate the CRLB for p and q

The CRLB is the inverse of the FIM, $\mathbf{C} = \mathbf{I}^{-1}$. Let $S(k) = \sum_{i=0}^{N-1} e^{-kx_i}$. The sums are geometric series where $S(k) = \frac{1 - e^{-kNdx}}{1 - e^{-kdx}}$.

The CRLB for p and q are:

- $\mathbf{CRLB}(p) = \sigma^2 \frac{S(2A + 2\Delta A)}{S(2A)S(2A + 2\Delta A) - S(2A + \Delta A)^2}$
- $\mathbf{CRLB}(q) = \sigma^2 \frac{S(2A)}{S(2A)S(2A + 2\Delta A) - S(2A + \Delta A)^2}$

3. Solve $\Delta A(A)$ for Constant CRLB

Let T=Ndx.

For large N we approximate the sum with an integral over the interval T:

$$S(k) = \frac{N}{T}\int_0^T e^{-kx}dx = \frac{N}{T}\cdot\frac{1-e^{-kT}}{k}$$

For *k=2AT*, the term exp(-*2AT*)<<1:
T=3500 (b-value range)
A=3.00x10^{-3} exp(-*2AT*)=7.5x10^{-10}
A=0.50x10^{-3} exp(-*2AT*)=0.030
A=0.25x10^{-3} exp(-*2AT*)=0.173 (!)
Thus $S(k) \approx \frac{N}{kT}$

$$\frac{\mathrm{CRLB}\ (p)}{\sigma^2}\cdot\frac{T}{N} \approx \frac{\frac{1}{2A+2\Delta A}}{\frac{1}{2A}\cdot\frac{1}{2A+2\Delta A}-\frac{1}{(2A+\Delta A)^2}} = \frac{2A(2A+\Delta A)^2}{\Delta A^2} = \mathrm{const}$$

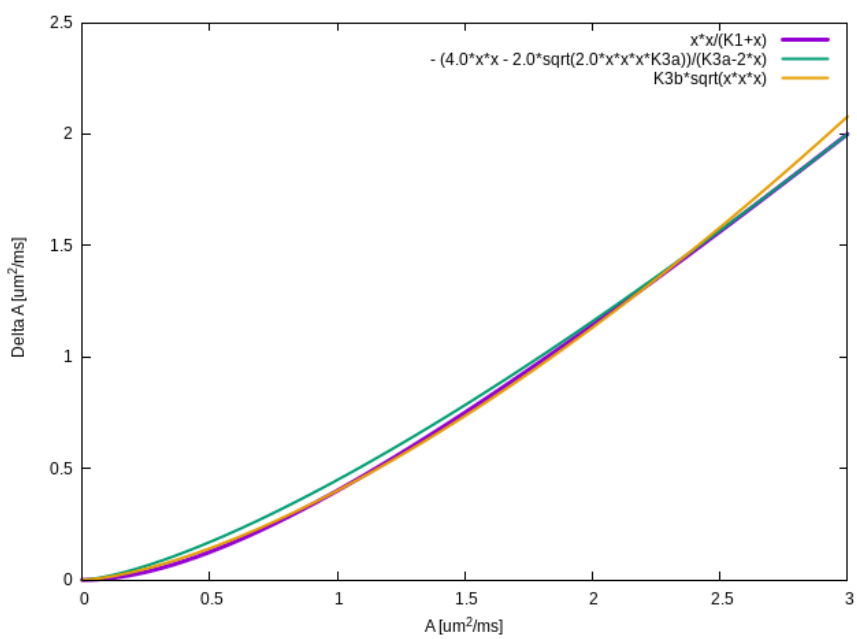
$$\Delta A = \frac{4A^2 \pm 2\sqrt{2A^3K}}{K-2A}$$

$$\frac{\mathrm{CRLB}\ (q)}{\sigma^2}\cdot\frac{T}{N} \approx \frac{\frac{1}{2A}}{\frac{1}{2A}\cdot\frac{1}{2A+2\Delta A}-\frac{1}{(2A+\Delta A)^2}} = \frac{2(A+\Delta A)(2A+\Delta A)^2}{\Delta A^2} = \mathrm{const}$$

For $\Delta A \ll A,\ A \approx A + \Delta A$

$$dA \approx \pm K A^{3/2}$$

Plot of $\Delta A(A)$ for solution 1 (equal interval), 3a (CRLB, for *dA<0*), and 3b (CRLB simplified, same as CRLB 2b)



CRLB (4 diffusion coefficients *A, B, C, D*)

For the function $f(x) = pe^{-Ax} + qe^{-Bx} + re^{-Cx} + se^{-Dx}$ with additive white Gaussian noise (AWGN) of variance σ^2, the **Cramér-Rao Lower Bound (CRLB)** for the parameters $\boldsymbol{\theta} = [p, q, r, s]^T$ is given by the diagonal elements of the inverse Fisher Information Matrix (FIM), $\mathbf{I}^{-1}(\boldsymbol{\theta})$.

Specifically, for N observations at points x_i, the CRLB for each parameter is:

$$\mathrm{Var}(\hat{p}) \geq [I(\boldsymbol{\theta})^{-1}]_{11}, \quad \mathrm{Var}(\hat{q}) \geq [I(\boldsymbol{\theta})^{-1}]_{22}, \quad \mathrm{Var}(\hat{r}) \geq [I(\boldsymbol{\theta})^{-1}]_{33}, \quad \mathrm{Var}(\hat{s}) \geq [I(\boldsymbol{\theta}$$

1. Define the Fisher Information Matrix (FIM)

For a signal in Gaussian noise, the elements of the FIM, I_{ij}, are calculated using the partial derivatives of the model function $f(x)$ with respect to the parameters. YouTube

- **Partial Derivatives**:
  - $\frac{\partial f}{\partial p} = e^{-Ax}$
  - $\frac{\partial f}{\partial q} = e^{-Bx}$
  - $\frac{\partial f}{\partial r} = e^{-Cx}$
  - $\frac{\partial f}{\partial s} = e^{-Dx}$
- **Matrix Elements**:

$$I_{ij} = \frac{1}{\sigma^2} \sum_{k=1}^{N} \left[\frac{\partial f(x_k)}{\partial \theta_i} \cdot \frac{\partial f(x_k)}{\partial \theta_j} \right]$$

2. Construct the 4×4 Matrix

The FIM for this multi-exponential model is a symmetric matrix where each entry represents the interaction between the exponential decay terms:

$$\mathbf{I} = \frac{1}{\sigma^2} \begin{bmatrix} \sum e^{-2Ax_k} & \sum e^{-(A+B)x_k} & \sum e^{-(A+C)x_k} & \sum e^{-(A+D)x_k} \\ \sum e^{-(B+A)x_k} & \sum e^{-2Bx_k} & \sum e^{-(B+C)x_k} & \sum e^{-(B+D)x_k} \\ \sum e^{-(C+A)x_k} & \sum e^{-(C+B)x_k} & \sum e^{-2Cx_k} & \sum e^{-(C+D)x_k} \\ \sum e^{-(D+A)x_k} & \sum e^{-(D+B)x_k} & \sum e^{-(D+C)x_k} & \sum e^{-2Dx_k} \end{bmatrix}$$

3. Calculate the Bounds

The CRLB for any individual parameter is found by inverting this matrix:

1. **Invert the Matrix**: Compute $\mathbf{C} = \mathbf{I}^{-1}$.
2. **Extract Variances**: The lower bound on the variance for each parameter is the corresponding diagonal element C_{ii}.
3. **Observation**: Because the basis functions ($e^{-Ax}, e^{-Bx}, \ldots$) are not orthogonal, the off-diagonal terms in the FIM are non-zero. This indicates **parameter coupling**, which generally increases the CRLB (minimum variance) compared to a single-component model.

✅ Answer

The CRLB for the parameters p, q, r, s is obtained from the diagonal of the inverse of the matrix $\mathbf{I}$, where $\mathbf{I}_{ij} = \frac{1}{\sigma^2} \sum_{k=1}^{N} \frac{\partial f(x_k)}{\partial \theta_i} \frac{\partial f(x_k)}{\partial \theta_j}$. Specifically, $\mathbf{Var}(\hat{\theta}_i) \geq [\mathbf{I}^{-1}]_{ii}$, which for this model depends heavily on the separation between the decay constants A, B, C, and D.

Let T=Ndx.

For large N we approximate the sum with an integral over the interval T:

$$S(k) = \frac{N}{T} \int_0^T e^{-kx} dx = \frac{N}{T} \cdot \frac{1 - e^{-kT}}{k}$$

We assume again the term e^{-kT}<<1. Thus $S(k) \approx \frac{N}{Tk}$

$$\begin{pmatrix} \frac{1}{6.00} & \frac{1}{5.00} & \frac{1}{4.50} & \frac{1}{4.00} & \frac{1}{3.75} & \frac{1}{3.50} & \frac{1}{3.25} \\ \frac{1}{5.00} & \frac{1}{4.00} & \frac{1}{3.50} & \frac{1}{3.00} & \frac{1}{2.75} & \frac{1}{2.50} & \frac{1}{2.25} \\ \frac{1}{4.50} & \frac{1}{3.50} & \frac{1}{3.00} & \frac{1}{2.50} & \frac{1}{2.25} & \frac{1}{2.00} & \frac{1}{1.75} \\ \frac{1}{4.00} & \frac{1}{3.00} & \frac{1}{2.50} & \frac{1}{2.00} & \frac{1}{1.75} & \frac{1}{1.50} & \frac{1}{1.25} \\ \frac{1}{3.75} & \frac{1}{2.75} & \frac{1}{2.25} & \frac{1}{1.75} & \frac{1}{1.50} & \frac{1}{1.25} & \frac{1}{1.00} \\ \frac{1}{3.50} & \frac{1}{2.50} & \frac{1}{2.00} & \frac{1}{1.50} & \frac{1}{1.25} & \frac{1}{1.00} & \frac{1}{0.75} \\ \frac{1}{3.25} & \frac{1}{2.25} & \frac{1}{1.75} & \frac{1}{1.25} & \frac{1}{1.00} & \frac{1}{0.75} & \frac{1}{0.50} \end{pmatrix}^{(-1)} = \begin{pmatrix} 41062.8 & -196560 & 291861.8 & -260590.9 & 152880 & -29781.8 & 1624.5 \\ -196560 & 980100 & -1496880 & 1386000 & -831600 & 166320 & -9360 \\ 291861.8 & -1496880 & 2333772 & -2222640 & 1358280 & -277830 & 16082.2 \\ -260590.9 & 1386000 & -2222640 & 2205000 & -1386000 & 294000 & -17869.1 \\ 152880 & -831600 & 1358280 & -1386000 & 889350 & -194040 & 12285 \\ -29781.8 & 166320 & -277830 & 294000 & -194040 & 44100 & -2978.2 \\ 1624.5 & -9360 & 16082.2 & -17869.1 & 12285 & -2978.2 & 226.3 \end{pmatrix}$$

Summary of Diagonal Value Square Roots (0 Digits Precision):

SNR: $\sqrt{\frac{N}{T\sigma^2}}$ / {203, 990, 1528, 1485, 943, 210, 15}

Assuming σ=1/300, T=3500/1000 s/mm^2, N=16

➔ SNR: {3.16, 0.65, 0.42, 0.43, 0.68, 3.05, 42.76} for D {3.00, 2.00, 1.50, 1.00, 0.75, 0.50, 0.25}

$$\begin{pmatrix} \frac{1}{40.00} & \frac{1}{23.00} & \frac{1}{22.00} & \frac{1}{21.50} & \frac{1}{21.00} & \frac{1}{20.75} & \frac{1}{20.50} & \frac{1}{20.25} \\ \frac{1}{23.00} & \frac{1}{6.00} & \frac{1}{5.00} & \frac{1}{4.50} & \frac{1}{4.00} & \frac{1}{3.75} & \frac{1}{3.50} & \frac{1}{3.25} \\ \frac{1}{22.00} & \frac{1}{5.00} & \frac{1}{4.00} & \frac{1}{3.50} & \frac{1}{3.00} & \frac{1}{2.75} & \frac{1}{2.50} & \frac{1}{2.25} \\ \frac{1}{21.50} & \frac{1}{4.50} & \frac{1}{3.50} & \frac{1}{3.00} & \frac{1}{2.50} & \frac{1}{2.25} & \frac{1}{2.00} & \frac{1}{1.75} \\ \frac{1}{21.00} & \frac{1}{4.00} & \frac{1}{3.00} & \frac{1}{2.50} & \frac{1}{2.00} & \frac{1}{1.75} & \frac{1}{1.50} & \frac{1}{1.25} \\ \frac{1}{20.75} & \frac{1}{3.75} & \frac{1}{2.75} & \frac{1}{2.25} & \frac{1}{1.75} & \frac{1}{1.50} & \frac{1}{1.25} & \frac{1}{1.00} \\ \frac{1}{20.50} & \frac{1}{3.50} & \frac{1}{2.50} & \frac{1}{2.00} & \frac{1}{1.50} & \frac{1}{1.25} & \frac{1}{1.00} & \frac{1}{0.75} \\ \frac{1}{20.25} & \frac{1}{3.25} & \frac{1}{2.25} & \frac{1}{1.75} & \frac{1}{1.25} & \frac{1}{1.00} & \frac{1}{0.75} & \frac{1}{0.50} \end{pmatrix}^{(-1)} =$$

$$\begin{pmatrix} 243.6 & -2882.3 & 10858.7 & -14119.0 & 10910.7 & -5922.9 & 1063.1 & -53.2 \\ -2882.3 & 75163.4 & -325030.6 & 458905.1 & -389676.2 & 222954.9 & -42359.4 & 2253.4 \\ 10858.7 & -325030.6 & 1464100 & -2126198.9 & 1872315.8 & -1095600 & 213704.6 & -11729.6 \\ -14119.0 & 458905.1 & -2126198.9 & 3152041.2 & -2854970.0 & 1701544.9 & -339441.6 & 19163.3 \\ 10910.7 & -389676.2 & 1872315.8 & -2854970.0 & 2693642.7 & -1651263.2 & 341611.3 & -20250.0 \\ -5922.9 & 222954.9 & -1095600 & 1701544.9 & -1651263.2 & 1033350 & -219886.2 & 13577.5 \\ 1063.1 & -42359.4 & 213704.6 & -339441.6 & 341611.3 & -219886.2 & 48739.1 & -3210.2 \\ -53.2 & 2253.4 & -11729.6 & 19163.3 & -20250.0 & 13577.5 & -3210.2 & 237.9 \end{pmatrix}$$

Summary of Diagonal Value Square Roots (0 Digits Precision):

SNR: $\sqrt{\frac{N}{T\sigma^2}}$ / {16, 274, 1210, 1775, 1641, 1017, 221, 15}

Assuming σ=1/300, T=3500/1000 s/mm^2, N=16

➔ SNR: {40.1, 2.34, 0.53, 0.36, 0.39, 0.63, 2.90, 42.8} for D {20.0, 3.00, 2.00, 1.50, 1.00, 0.75, 0.50, 0.25}

$$\begin{pmatrix} \frac{1}{6.00} & \frac{1}{4.00} & \frac{1}{3.60} & \frac{1}{3.43} & \frac{1}{3.33} & \frac{1}{3.27} & \frac{1}{3.23} \\ \frac{1}{4.00} & \frac{1}{2.00} & \frac{1}{1.60} & \frac{1}{1.43} & \frac{1}{1.33} & \frac{1}{1.27} & \frac{1}{1.23} \\ \frac{1}{3.60} & \frac{1}{1.60} & \frac{1}{1.20} & \frac{1}{1.03} & \frac{1}{0.93} & \frac{1}{0.87} & \frac{1}{0.83} \\ \frac{1}{3.43} & \frac{1}{1.43} & \frac{1}{1.03} & \frac{1}{0.86} & \frac{1}{0.76} & \frac{1}{0.70} & \frac{1}{0.66} \\ \frac{1}{3.33} & \frac{1}{1.33} & \frac{1}{0.93} & \frac{1}{0.76} & \frac{1}{0.66} & \frac{1}{0.60} & \frac{1}{0.56} \\ \frac{1}{3.27} & \frac{1}{1.27} & \frac{1}{0.87} & \frac{1}{0.70} & \frac{1}{0.60} & \frac{1}{0.54} & \frac{1}{0.50} \\ \frac{1}{3.23} & \frac{1}{1.23} & \frac{1}{0.83} & \frac{1}{0.66} & \frac{1}{0.56} & \frac{1}{0.50} & \frac{1}{0.46} \end{pmatrix}^{(-1)} = \begin{pmatrix} 291.9 & -2316.7 & 10329.8 & -23355.8 & 30102.6 & -20762.5 & 5752.5 \\ -2316.7 & 24517.5 & -122985.8 & 296436.5 & -398819.3 & 282880.3 & -79934.0 \\ 10329.8 & -122985.8 & 658054.8 & -1651577.5 & 2288828.6 & -1657126.0 & 475364.7 \\ -23355.8 & 296436.5 & -1651577.5 & 4261187.0 & -6033585.6 & 4436797.3 & -1287816.3 \\ 30102.6 & -398819.3 & 2288828.6 & -6033585.6 & 8693710.3 & -6477037.5 & 1899196.2 \\ -20762.5 & 282880.3 & -1657126.0 & 4436797.3 & -6477037.5 & 4874302.3 & -1440676.3 \\ 5752.5 & -79934.0 & 475364.7 & -1287816.3 & 1899196.2 & -1440676.3 & 428557.2 \end{pmatrix}$$

Summary of Diagonal Value Square Roots (0 Digits Precision):

SNR: $\sqrt{\frac{N}{T\sigma^2}}$ / {18, 156, 811, 2065, 2955, 2209, 655}

Assuming σ=1/300, T=3500/1000 s/mm^2, N=16

➔ SNR: {35.63, 4.11, 0.79, 0.31, 0.22, 0.29, 0.98} for D {3.00, 1.00, 0.60, 0.43, 0.33, 0.27, 0.23}

$$\begin{pmatrix} \frac{1}{40.00} & \frac{1}{23.00} & \frac{1}{21.00} & \frac{1}{20.60} & \frac{1}{20.43} & \frac{1}{20.33} & \frac{1}{20.27} & \frac{1}{20.23} \\ \frac{1}{23.00} & \frac{1}{6.00} & \frac{1}{4.00} & \frac{1}{3.60} & \frac{1}{3.43} & \frac{1}{3.33} & \frac{1}{3.27} & \frac{1}{3.23} \\ \frac{1}{21.00} & \frac{1}{4.00} & \frac{1}{2.00} & \frac{1}{1.60} & \frac{1}{1.43} & \frac{1}{1.33} & \frac{1}{1.27} & \frac{1}{1.23} \\ \frac{1}{20.60} & \frac{1}{3.60} & \frac{1}{1.60} & \frac{1}{1.20} & \frac{1}{1.03} & \frac{1}{0.93} & \frac{1}{0.87} & \frac{1}{0.83} \\ \frac{1}{20.43} & \frac{1}{3.43} & \frac{1}{1.43} & \frac{1}{1.03} & \frac{1}{0.86} & \frac{1}{0.76} & \frac{1}{0.70} & \frac{1}{0.66} \\ \frac{1}{20.33} & \frac{1}{3.33} & \frac{1}{1.33} & \frac{1}{0.93} & \frac{1}{0.76} & \frac{1}{0.66} & \frac{1}{0.60} & \frac{1}{0.56} \\ \frac{1}{20.27} & \frac{1}{3.27} & \frac{1}{1.27} & \frac{1}{0.87} & \frac{1}{0.70} & \frac{1}{0.60} & \frac{1}{0.54} & \frac{1}{0.50} \\ \frac{1}{20.23} & \frac{1}{3.23} & \frac{1}{1.23} & \frac{1}{0.83} & \frac{1}{0.66} & \frac{1}{0.56} & \frac{1}{0.50} & \frac{1}{0.46} \end{pmatrix}^{(-1)} =$$

$$\begin{pmatrix} 129.8 & -177.3 & 839.6 & -3300.0 & 7047.3 & -8773.4 & 5924.1 & -1618.0 \\ -177.3 & 534.3 & -3464.3 & 14840.1 & -32987.6 & 42093.6 & -28859.3 & 7963.9 \\ 839.6 & -3464.3 & 29950.8 & -144339.8 & 342038.5 & -455590.7 & 321214.4 & -90403.7 \\ -3300.0 & 14840.1 & -144339.8 & 741981.5 & -1830804.6 & 2511954.5 & -1807788.7 & 516513.4 \\ 7047.3 & -32987.6 & 342038.5 & -1830804.6 & 4643930.1 & -6510075.7 & 4758540.8 & -1375690.3 \\ -8773.4 & 42093.6 & -455590.7 & 2511954.5 & -6510075.7 & 9286909.2 & -6877587.0 & 2008593.6 \\ 5924.1 & -28859.3 & 321214.4 & -1807788.7 & 4758540.8 & -6877587.0 & 5144767.9 & -1514545.4 \\ -1618.0 & 7963.9 & -90403.7 & 516513.4 & -1375690.3 & 2008593.6 & -1514545.4 & 448732.2 \end{pmatrix}$$

Summary of Diagonal Value Square Roots (0 Digits Precision):

SNR: $\sqrt{\frac{N}{T\sigma^2}}$ / {11, 23, 173, 861, 2155, 3047, 2268, 670}

Assuming σ=1/300, T=3500/1000 s/mm^2, N=16

➔ SNR: {58.31, 27.89, 3.71, 0.75, 0.30, 0.21, 0.28, 0.96} for D {20.0, 3.00, 1.00, 0.60, 0.43, 0.33, 0.27, 0.23}